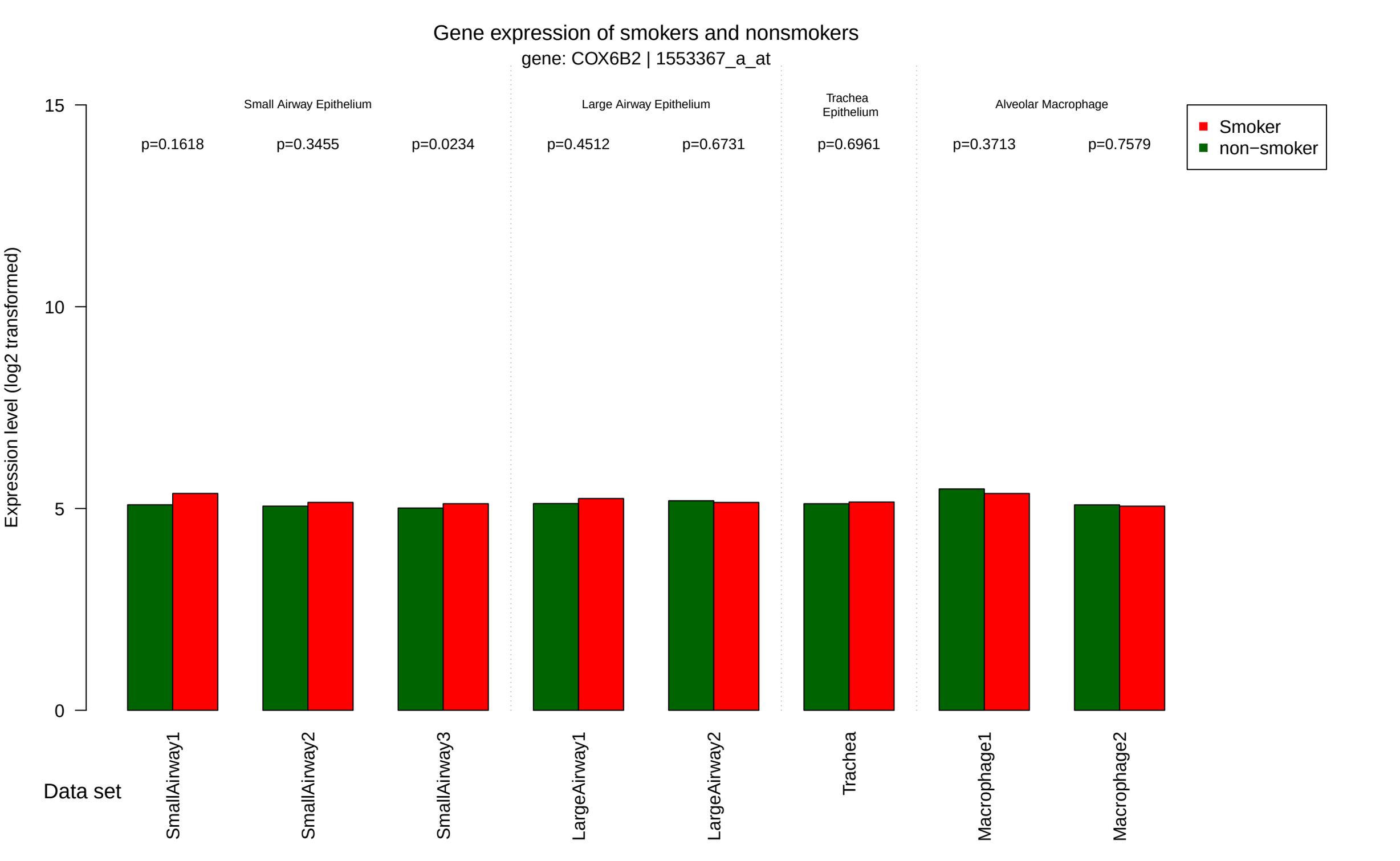
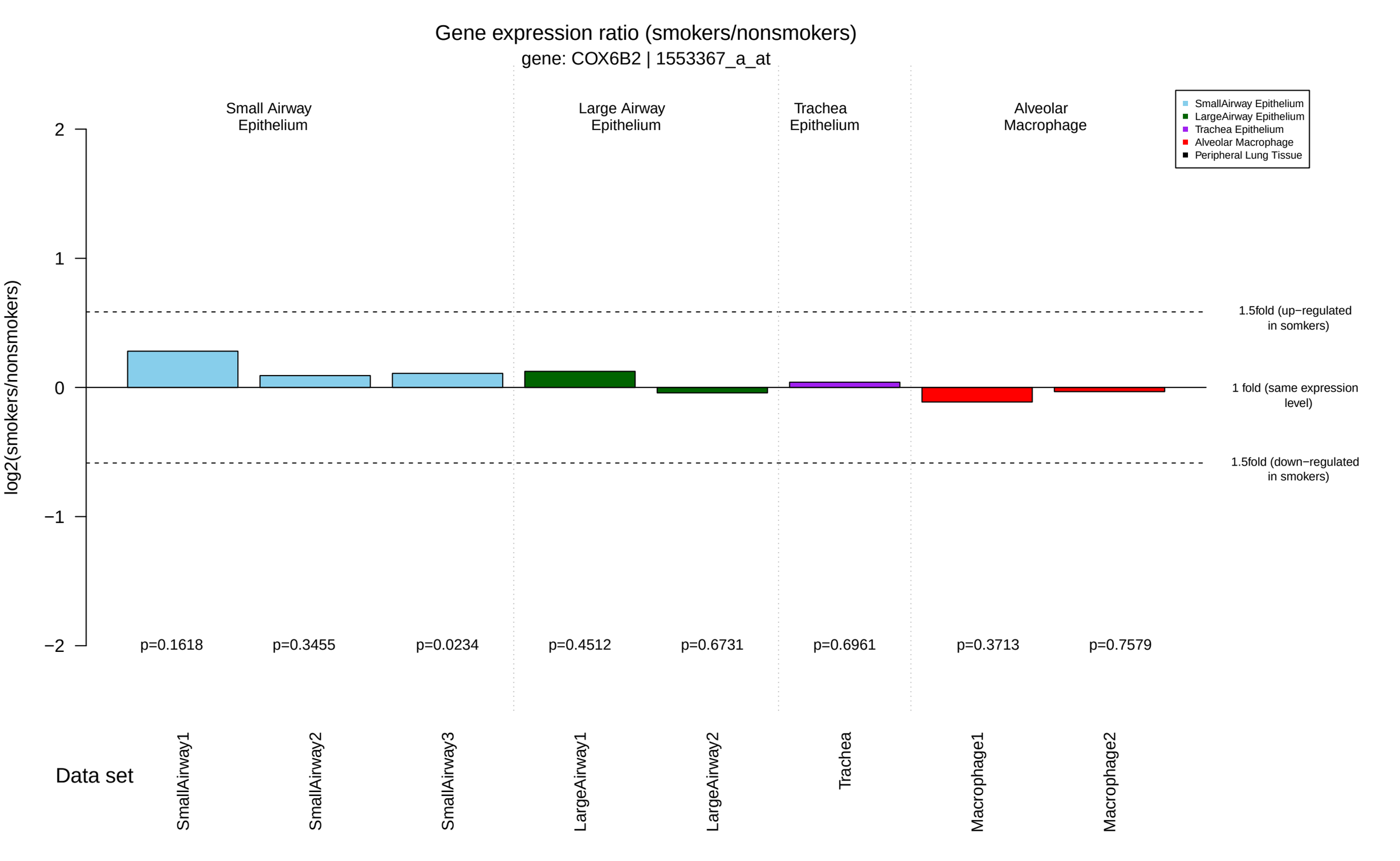
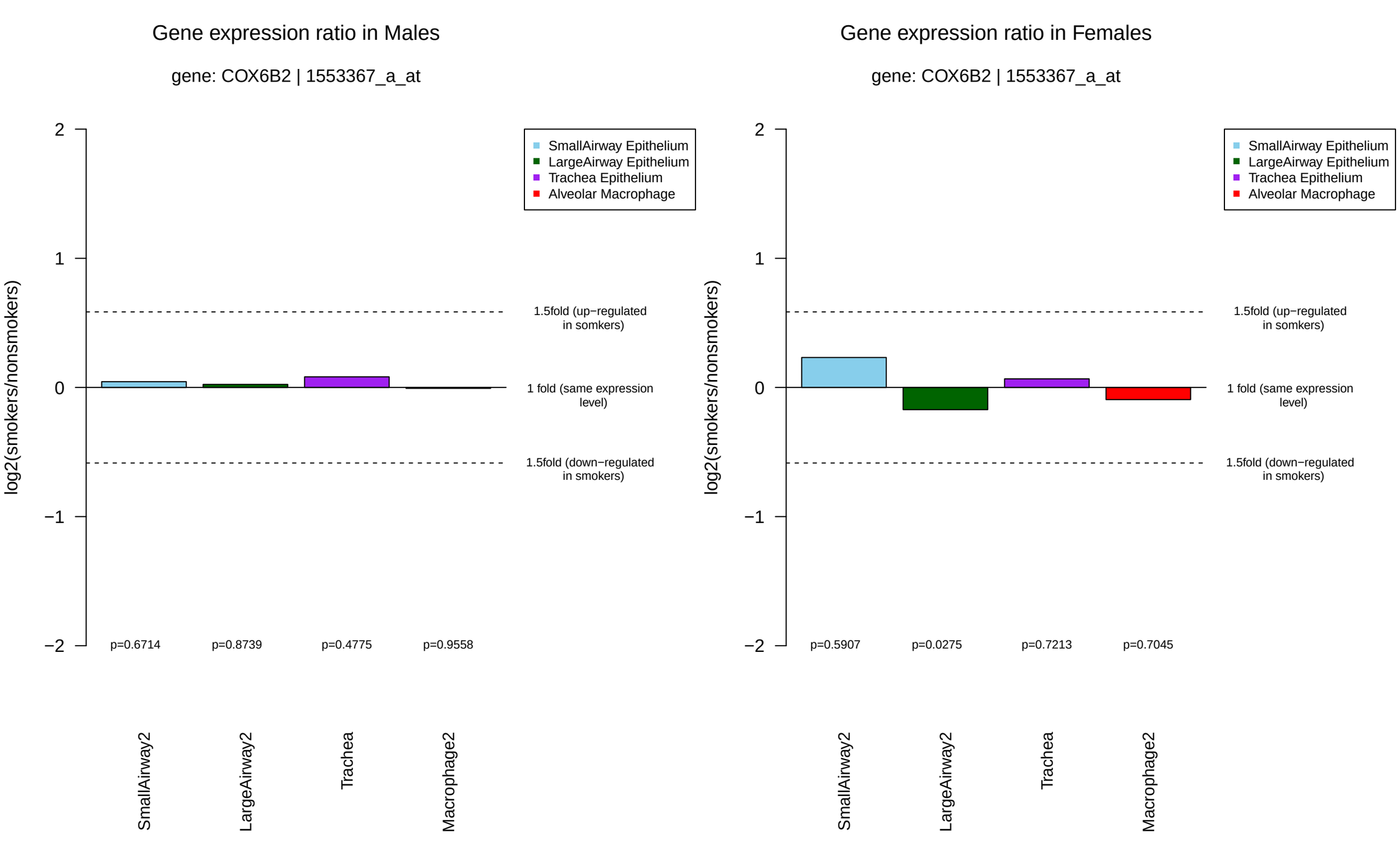
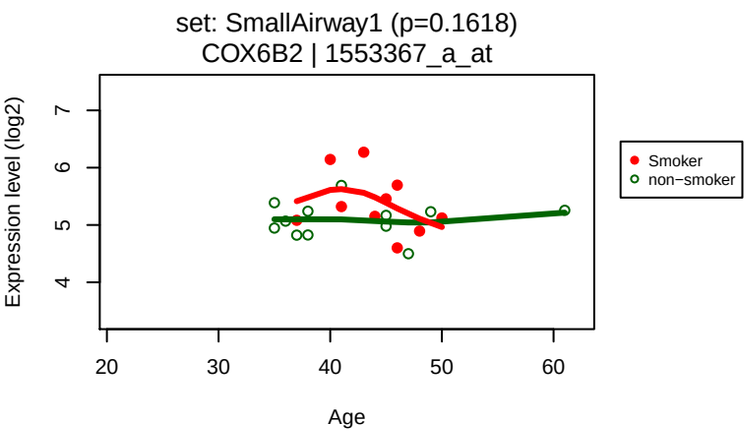
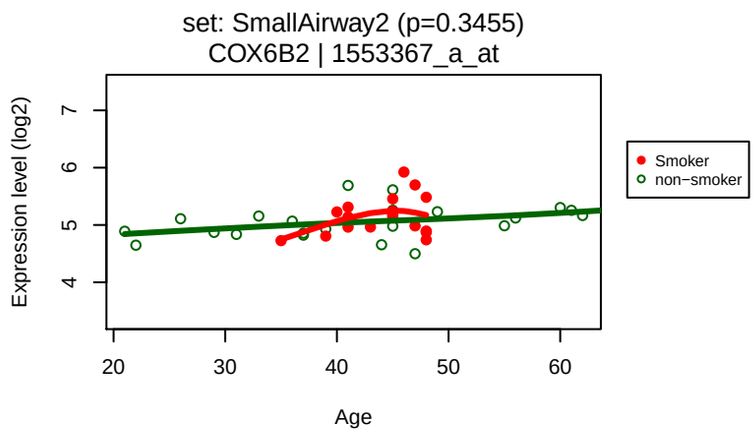
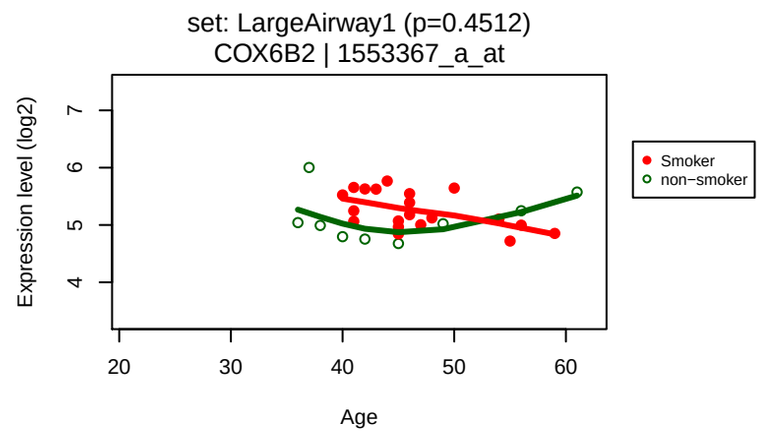

Gene expression of smokers and nonsmokers
gene: COX6B2 | 1553367_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1618
p=0.3455
p=0.0234
p=0.4512
p=0.6731
p=0.6961
p=0.3713
p=0.7579
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

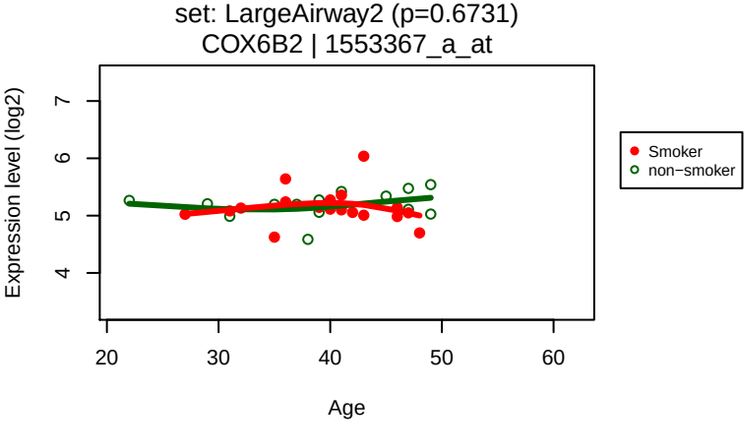

Gene expression ratio (smokers/nonsmokers)
gene: COX6B2 | 1553367_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1618
p=0.3455
p=0.0234
p=0.4512
p=0.6731
p=0.6961
p=0.3713
p=0.7579
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

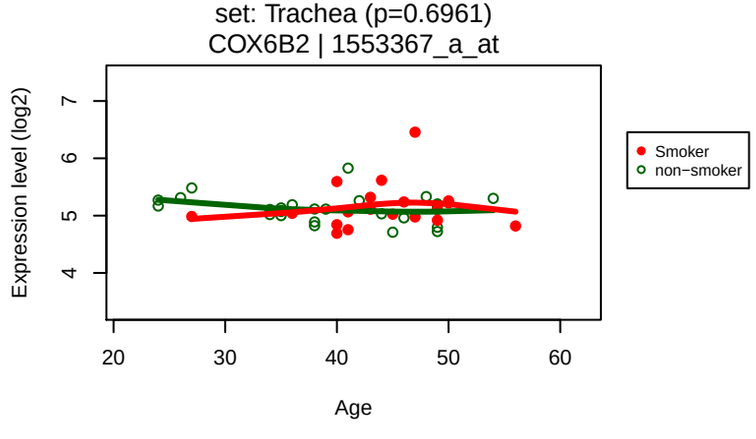

Gene expression ratio in Males
gene: COX6B2 | 1553367_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6714
p=0.8739
p=0.4775
p=0.9558
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COX6B2 | 1553367_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5907
p=0.0275
p=0.7213
p=0.7045
SmallAirway2
LargeAirway2
Trachea
Macrophage2

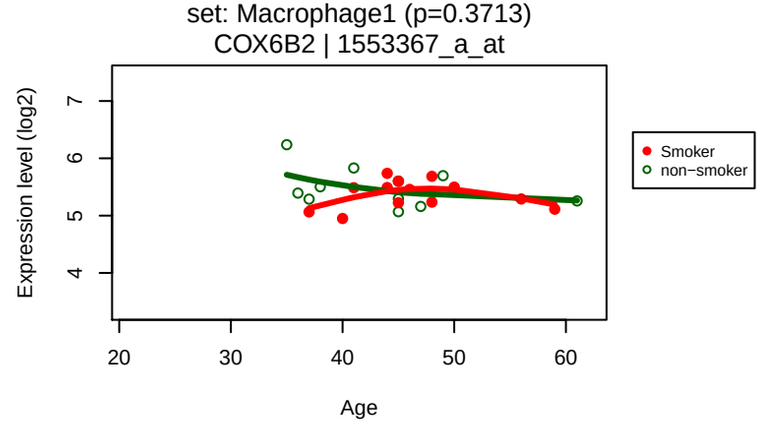

set: SmallAirway1 (p=0.1618)
COX6B2 | 1553367_a_at
Expression level (log2)
Age
Smoker
non−smoker

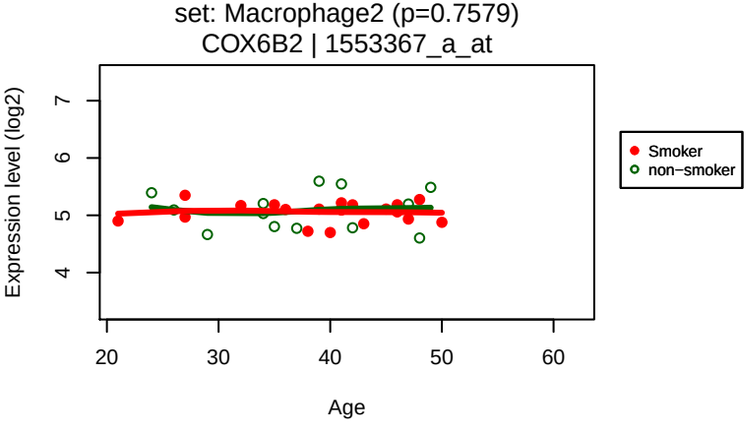

set: SmallAirway2 (p=0.3455)
COX6B2 | 1553367_a_at
Expression level (log2)
Age
Smoker
non−smoker

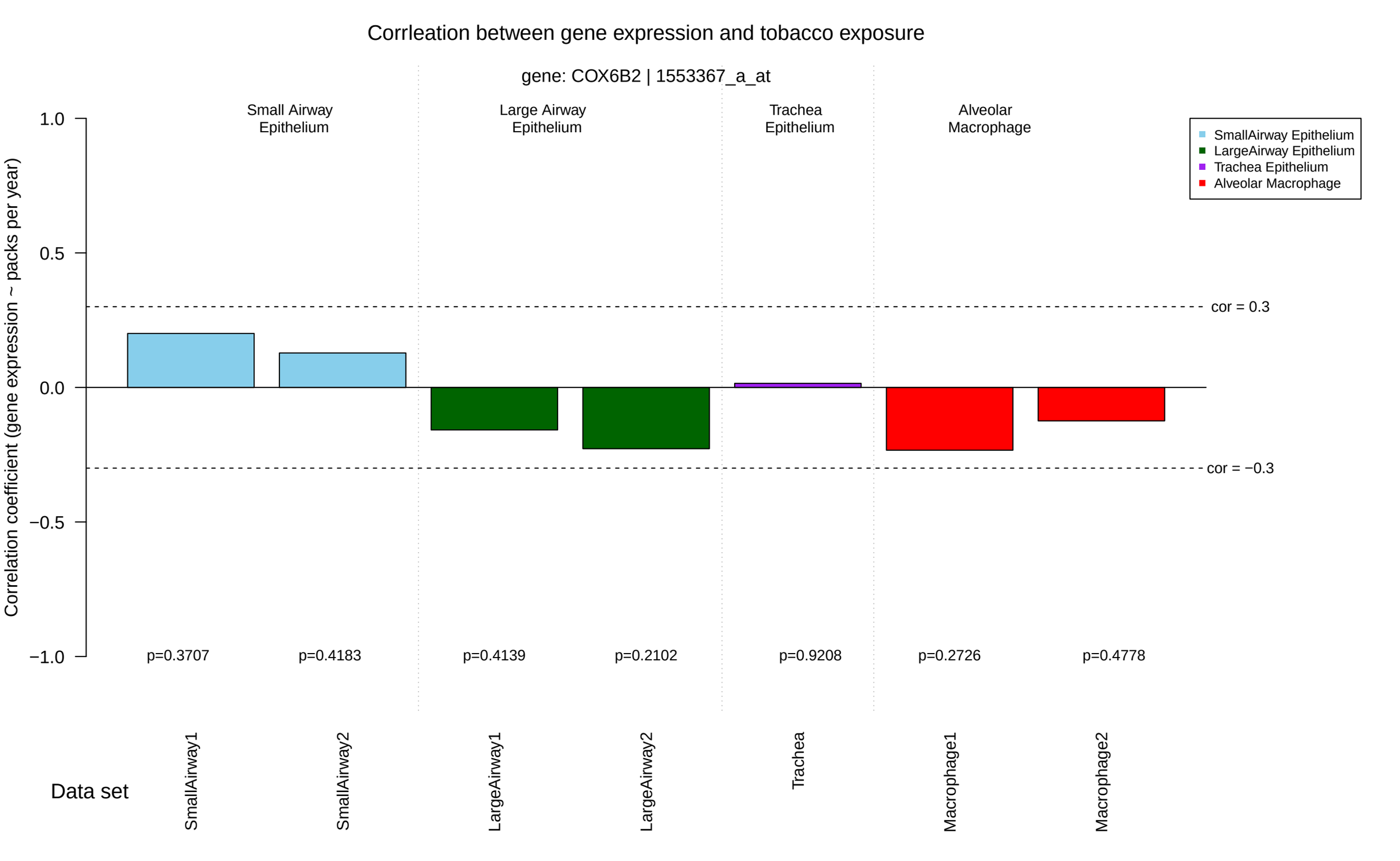

set: LargeAirway1 (p=0.4512)
COX6B2 | 1553367_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6731)
COX6B2 | 1553367_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6961)
COX6B2 | 1553367_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3713)
COX6B2 | 1553367_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7579)
COX6B2 | 1553367_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COX6B2 | 1553367_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3707
p=0.4183
p=0.4139
p=0.2102
p=0.9208
p=0.2726
p=0.4778
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2